

DeepFear: Game Usage within Virtual Reality to Provoke Physiological Responses of Fear

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ABSTRACT

The investigation and the classification of the physiological signals involved in fear perception is complicated due to the difficulties in reliably eliciting and measuring the complex construct of fear. Especially, using Virtual Reality (VR) games can well elicit the physiological responses, then it can be used developing treatments in healthcare domain. In this study, we carried out exploratory physiological data analysis and wearable sensory device feasibility for the responses of fear. We contributed 1) to use a of-the-shelf commercial game (Half Life-Alyx) to provoke fear emotion, 2) to demonstrate a performance analysis with different deep learning models like Convolutional Neural Network (CNN), Long-Short Term Memory (LSTM) and Transformer, 3) to investigate the most responsive physiological signal by comprehensive data analysis and best sensory device in terms of multi-level of fear classification. Accuracy metrics, f1-scores and confusion matrices showed that ECG and ACC are the most significant two signals for fear recognition.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**.

KEYWORDS

Fear Classification, Virtual Reality, Physiological Data, Game, Machine Learning, Deep Learning

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1 INTRODUCTION

VR is currently employed and widely utilized across various application domains. Previous studies have suggested that VR technology has the capability to evoke more intense emotions compared to less immersive technologies like video and audio presentations [26]. Specifically, the elicitation of fear emotions in horror and survival

games remains underexplored in terms of physiological signal response. Uncovering significant and responsive physiological data for fear recognition is crucial for treatments such as anxiety disorders [6, 15] and phobias [1], particularly with the use of unobtrusive wearable sensory devices.

However, within the realm of emotion recognition, fear recognition has not received adequate attention when compared to other emotional stimuli. Furthermore, traditional machine learning approaches have proven insufficient in delivering satisfactory results [8, 28, 30]. The emergence of deep learning approaches has paved the way for effectively addressing this task, leveraging the abundance of available data [2, 14, 19].

This paper aims to explore fear responses through physiological data collected by wearable sensory devices during the gameplay of a survival-horror commercial game, with a specific focus on implementing deep learning techniques. Our investigation delves into the feasibility of utilizing different physiological data for fear recognition, examining its correlation with subjective feedback, and demonstrating classification capabilities for this purpose. In doing so, we contribute to the broader understanding of how deep learning can enhance the analysis of fear responses in virtual reality environments.

2 RELATED WORK

2.1 Games for Provoking Fear Stimuli

Fear is a multifaceted psychological phenomenon involving both cognitive and physiological elements, triggered by the perception of a stressful stimulus through sensory organs such as the eyes, ears, and skin. This experience results in the release of chemical substances, provoking physiological responses like an increased heart rate, accelerated breathing, and heightened muscular activation characteristic of the fight-or-flight response.

Virtual Reality (VR) games, particularly those in the horror genre, have demonstrated effectiveness in eliciting fear responses from players due to the immersive and realistic experiences they provide [13, 21]. Various game design techniques are employed, including restricting movement, visibility through darkness, and obscured environments [17], along with jump scares, realistic graphics, and spatial audio to induce fear [25]. Consequently, VR is expected to create a strong sense of spatial presence in a frightening virtual environment, intensifying the perception that potential monsters are genuinely present, thereby heightening feelings of fear. Moreover, studies have indicated that virtual reality can evoke emotional and behavioral responses closely resembling those experienced in real-life scenarios [33, 34, 39].

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The assessment of the occurrence and intensity of fear often relies on subjective self-reports. Lemmens et al. [21] used Resident Evil 7 [9] in their study to induce fear, employing statistical features such as mean and standard deviation to estimate the correlation between fear and loathing using a presence questionnaire and the State Hostility Scale [3]. Zhang et al. [46] explored gender-specific user experiences and the efficacy of VR horror games using the PANAS-X [41] questionnaire. In contrast, alternative approaches to measuring fear involve physiological and behavioral data, such as heart rate and skin conductance. However, studies on players' experiences and emotional responses while playing VR games, especially in the context of commercial survival-horror games and fear elicitation, are still limited.

2.2 Fear Perception and Classification

In the existing literature, studies commonly employ a traditional machine learning approach, utilizing questionnaire results as subjective feedback [29]. The predominant method for fear classification involves interpreting valence and arousal [40]. The utilization of physiological data for fear recognition tasks, particularly through the implementation of deep learning models, is relatively limited. Fu et al. [13], for example, developed a dataset incorporating body gestures as participants engaged in VR horror games. They employed a bidirectional LSTM network and achieved an accuracy of 71.43% with cross-validation. However, their performance in fear label classification was notably inferior compared to non-fear labels. In a different study, Luong et al. [22] examined short-term and momentary physiological responses to fear using SVM, Logistic Regression and Random Forest, aiming for a binary classification of fear and frustration. Zhan et al. [45] adopted a zero-shot learning approach, classifying emotions through a CNN model and combining visual and affective embeddings as features. Some researchers have utilized EEG devices for fear classification tasks employing CNN networks, with accuracy being the primary metric [31]. However, EEG studies are characterized by complexity, time consumption, and limitations on participant movement.

The Transformer architecture, which relies solely on self-attention mechanisms, revolutionized machine learning by eliminating the need for recurrent layers [36]. Its success in various fields like language translation and sentiment analysis highlights its potential for time-series signal applications. Despite Transformer's success, there is limited literature on utilizing Transformer models with physiological data for fear classification. Some studies explored Transformer variants for emotion detection in physiological signals [37, 42, 44]. Kumar et al. [18] applied a Transformer-based model for emotion detection using context-dependent features from text.

3 MEASUREMENT DESIGN

3.1 Half Life-Alyx Game

"Half Life-Alyx" [35] a VR prequel to the popular series by Valve Corporation, isn't strictly a horror game but incorporates unsettling elements, especially in VR. Detailed graphics, well-designed environments, suspenseful pacing, and encounters with terrifying creatures like head-crabs contribute to a fearful atmosphere. The game also employs sophisticated sound design with ambient noises

and environmental hazards, intensifying the sense of vulnerability and fear in players.

3.2 Measurement Devices

To measure the participants' physiological signals during their VR game experience, we used three different devices. Since nowadays wearable sensors offer superior practicability with respect to cost, ease to use and portability, we selected a Polar H10 (Polar Electro Oy, Finland) sensory device, which is an electrode based chest strap, and Empatica E4 (Empatica Inc., USA) device, which is a medical grade wristband. Both of these devices transmit the data to the computer via Bluetooth communication. For game, we used HTC Vive Pro as headset (HMD) to retrieve the eye-tracking data.

3.3 Data Collection

The study involved 34 participants (14 female, 20 male), aged between 21 and 33 years, who willingly consented to participate in the experiments. Only two participants have played this game before. Participants donned sensory devices, and connections were established between the sensors and the measurement engine—a customized software developed in the Python programming language. Data streaming started simultaneously for all sensors (see Figure 1). Firstly, we collected 3 mins data as baseline. In this period of time, participants were freely moved with the sensors and selected the respective game chapter in the game menu. Then, prior to engaging in the game, participants completed PANAS-X[41] questionnaire to measure participants' positive and negative emotions during baseline data collection.

Throughout the study, an instructor was present to monitor participants in both the real and virtual worlds, ensuring continuous gameplay. All participants initiated their first session in "Chapter 1" and were then allowed to continue playing the game without additional instructions. After approximately 45 minutes, the session concluded, irrespective of the participant's progress in the game at that point. The same procedure was repeated for the second session on a different day, but each participant's starting point was changed to "Chapter 3." Right after the experiment, participants completed again PANAS-X questionnaire regarding specific sections of the two chapters to assess the level of discrete emotions. Additionally, participants identified and declared which parts and moments of the game induced fear for them. Despite the dataset encompassing both body motions and physiological data (as discussed in "Who is Alyx," [27]), this study specifically focused on utilizing processed physiological data.

The recorded data types are summarized in Table 1: Electrocardiography (ECG) and Acceleration (ACC) data using the Polar H10 chest strap; Photoplethysmography (PPG), ACC, Electrodermal Activity (EDA), Inter-beat Interval (IBI), and Peripheral Body Temperature (TEMP) data using the Empatica E4 wristband; and eye tracking (ET) data using the HTC Vive Pro HMD. Table 1 provides an overview of the data types, sampling rates, and the number of features extracted from the physiological data.

3.4 Labelling the Dataset

For data labelling, we used two different sections of the game for two different chapters: Train Compartment section in Chapter 1



Figure 1: First column shows game scenes from Train Compartment-Chapter 1, second column shows the game scenes from Dark Tunnel-Chapter 3, the last column shows an equipped participant and the respective sensors used during the experiments in detail.

and Dark Tunnel section in Chapter 3. After completing these chapters, each participant rated their discrete emotions with 60 items of PANAS-X [41] questionnaire. Since we focus on only fear emotion, we considered fear dimension of this questionnaire which consist of 6 items which effects fear emotion: afraid, scared, frightened, nervous, jittery, shaky. The scales for each item is between 1 and 5. Here, 1 refers very slightly or not at all and 5 refers extremely feeling of emotions. We calculated mean (M) and standard deviation (SD) of the total scores before playing the game as baseline (M=1.35, SD=0.65) and in game scores (M=3.83, SD= 1.04). We constructed dataset with three labels (no-low, medium and high) using participants' mean scores of 6 fear items by selecting the fear label scales following rule : 0-1.5 no-low level fear, 1.5-3.5 medium level fear, 3.5-5 high level of fear. The gameplay statistics for Train Compartment is (M = 4min 48s, SD = 1min 15s), for Train Compartment is (M = 4min 27s, SD = 1min 32s).

3.5 Data Processing and Feature Extraction

After raw data acquisition, the collected data was pre-processed in order to apply deep learning algorithms. Instead of using only raw data, we calculated pre-features as input to the models. Firstly, the data was normalized. For the ET (pupil diameters, gaze directions) and ACC (x,y,z axis) data, Z-score normalization (D_z) was used. Here, for each data sample D_i , the Z-score normalization can be calculated with the mean of the training samples D_u and the standard deviation of the training samples D_s . The new data sample value D_z replaces the old sample i . This standard technique guarantees the model to learn more smoothly on the data, since it is standard normally distributed (Equation 1).

The normalization of the BVP, EDA, IBI, TEMP and ECG data was done using a min-max scaler (D_{minmax}) which sets the values in relation to their max and min values which can vary for each person. Here, D_{min} and D_{max} refer to the minimum and maximum of the data samples per person, respectively. By using rolling moving average (D_{rma}), the features were extracted from the normalized data collected by Empatica and H10 sensors. This technique smooths the

data by taking the average value over the last n samples, instead of the raw signal. Given a fixed sampling rate, n corresponds a time window. According to [11], since an appropriate time for recognizing a change in physiological signals is 3 seconds. Depending on the sampling rate, this value defines the final choice for the parameter n . Hence, $n = 3 \times \text{sampling rate}$ (Equation 2).

For another feature class, the percentage of change (D_{pc}) was calculated from the normalized values. It indicates how much the value has proportionally changed in the number of n time steps, rather than computing a nominal difference (Equation 3). The last two features that were constructed from the normalized data are the maximum (max) and the minimum (min) in the last n time steps (Equation 4). As additional data source, the heart rate (HR) data was calculated from ECG data by using the algorithm in [10]. This technique detects the current beat by leveraging specified thresholds and R-R interval analysis. Aforementioned four features (rma, pc, min, max) were also calculated for HR data.

$$D_z = \frac{D_i - D_u}{D_s} \quad (1)$$

$$D_{minmax} = \frac{D_i - D_{min}}{D_{max} - D_{min}} \quad (2)$$

$$D_{rma} = \frac{1}{n} \sum_{z=0}^n D_z \quad (3)$$

$$D_{pc} = \frac{D_z - D_{z-n}}{D_{z-n}} \quad (4)$$

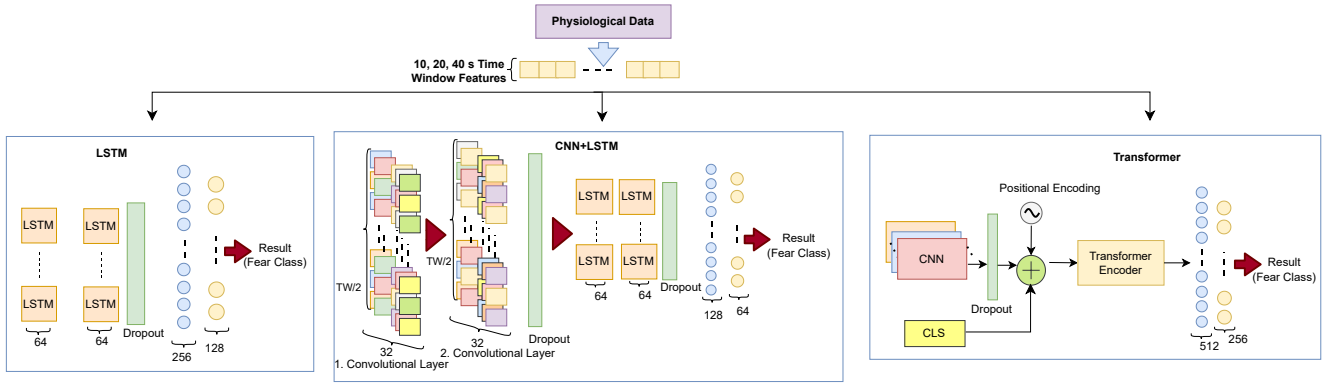
4 DEEP LEARNING MODELS AND TRAINING

We implemented four different from-scratch deep learning models for fear classification task. The overview of the models can be seen in Figure 2.

LSTM Model: The model consists of five layers: two LSTM layers, a dropout layer and two dense (fully-connected) layers. The input for the LSTM layer is a tensor of shape (batch size, time window, features). Each LSTM module produces a tensor of shape

Table 1: The features that are extracted and pre-processed from the raw sensory data to train the fear level classifier.

Device Type	S. Rate (Hz)	Data Type	Features
Eye-Tracker	80 Hz	Pupil Diameter ((left,right eye)	2
	80 Hz	Gaze direction(left,right eye; x,y,z values)	6
Empatica	32	ACC (x,y,z values; rma,pc,max,min)	12
	64	BVP (rma,pc,max,min)	4
	4	EDA (rma,pc,max,min)	4
	4	IBI (rma,pc,max,min)	4
	4	TEMP (rma,pc,max,min)	4
Polar H10	200 Hz	ACC (x,y,z values; rma,pc,max min)	12
	130 Hz	ECG (rma, pc, max, min)	4
	1	HR (rma, pc, max, min)	4

**Figure 2: The overview of the from-scratch LSTM, CNN-LSTM and Transformer models.**

(batch size, LSTM hidden size) as output, which contains the final hidden states of the input sequence after the last time step. After applying dropout, this output tensor is fed into the two dense layers. ReLU [24] activation function was used for the dense layers.

Beside that 4 layered LSTM Network, we also used bidirectional LSTM Network (Bid-LSTM) [32] with same LSTM architecture for the classification task. Standard LSTM network have restrictions as the future input information cannot be reached from the current state. On the contrary, Bid-LSTM networks do not require its input data as the same dimension. Moreover, their future input information can be reached from the current state. The main idea of Bid-LSTM is to connect two hidden layers of opposite directions to the same output. By this structure, the output layer can access information from past and future states and interpret them better.

CNN-LSTM Model: To improve the classification performance, we also deployed a CNN-LSTM architecture to acquire the spatial features and time-invariant patterns. This model consists of ten layers: two 1D convolution layers (Conv1D), two pooling layers, a dropout layer, followed by the five layers that were also present in the LSTM-model (two LSTM layers, a dropout layer and two dense layers). The input tensor for the first Conv1D layer is of shape (batch size, timesteps, features). Then, two 1D convolutions are applied. For both Conv1D layers, the number of filters is equal to the input size for the LSTM layer. The kernel size is 4, and ReLU is

applied as an activation function. After each Conv1D layers, max pooling is used in the pooling layer, with a pool size of 2 and a stride of 2. After the max pooling function, the output tensor is of shape (batch size, reduced timesteps, LSTM input size) and can be used as an input for the LSTM layer. The following LSTM and dense layers are set up in a similar way to the LSTM model described previously but with different size of dense layers.

Transformer Model: To process physiological data with Transformer model, we encoded the data into s feature vectors that represents each data samples of with d dimension. After encoding, we obtained the features $F = f_1, \dots, f_s$ where $f_i \in R^d$. Then, similar to BERT approach [12], at the output of the Transformer, we obtain an embedding of the classification token ($eCLS$), along with the rest of the representations of the signal. Through the attention mechanisms of the Transformer, $eCLS$ is capable of aggregating information from the entire input signal and its contextualized representations. To take into account the actual ordering of timesteps in the input sequence, positional information of each input is required for the Transformer. We sum the positional embeddings with the features F , and get $Z = CLS + pe_0, f_1 + pe_1, \dots, f_s + pe_s$, where $pe_i \in R^d$ is the positional embedding for time-step i . After normalization of Z [5], using Transformer encoder, we get contextualized representations E with h heads and l layers, $Transformer_{h \times l}(Z) = C = eCLS, e_1, \dots, e_s$. Then, we use C for fear level classification.

Table 2: The variables and their values that were used in the grid search to optimize the models’ hyperparameters and best performing values for the respective model type.

Model Type	Values	LSTM	Bid-LSTM	CNN+LSTM	Transformer
Time Window	10,20,40	30	30	30	50
LSTM Hidden	32, 64	64	64	64	-
Fully-Connected Layer	64, 128, 256, 512	256,128	256,128	128,64	512,256
Conv. Filters	8, 16	-	-	8	-
Learning Rate	0.001, 0.005	0.001	0.001	0.005	0.001
Dropout	0.5, 0.7	0.5	0.5	0.5	0.7
Epoch	Early Stop	140	140	260	120

The input encoder was built with 3 layers of 1D-CNN with ReLU [43] activation function. We used layer normalization [5] on the first layer, and at the output of the encoder. Kernel sizes are (65, 33, 17), the numbers of channels are (64, 128, 256) and the stride for all layers is 1. The Transformer in our signal encoder has a model dimension $d_{model} = 256$, 2 layers and 2 attention heads, with its FCN size of $d_{model} \cdot 2 = 512$. The FCN used to predict the masked values consists of a single linear layer of size $d_{model}/2 = 128$ followed by a ReLU activation function. An additional linear layer was used to project the output vector to a single value, which corresponds to the predicted value of a masked point.

4.1 Hyperparameter Optimization and Model Training

To ensure reliable performance results, we implemented 34-fold cross-validation for the dataset by splitting the subjects into 34 groups, therefore, each participant is in one group and called leave-one-subject-out cross-validation (LOSO). We trained the model on 33 subjects and tested it on one subject with for each iteration and consciously selected different persons for the test set to investigate the generalization capability of each model on never seen participants by the model. Additionally, a time window analysis was conducted by segmenting the data into 10, 20, and 40-second intervals to explore the temporal significance of deep features on the learning capacity of the models. This entire process involved a total of 6290 timesteps, comprising 2430 instances of no-low, 1731 medium, and 2129 high-level fear classes, thereby indicating a balanced distribution within our dataset. To find optimal hyperparameters for the each model type, we conducted a grid search covering different model configurations resulting in 1600 total training of models. The best hyperparameters that led to the best classification results are shown in Table 2. Models were trained on a machine with an Intel Core i7 9700K CPU, 32 GB RAM and NVIDIA RTX 2070 Super GPU with PyTorch 1.10 deep learning library.

5 RESULTS AND DISCUSSION

After training, we initially applied classification models to individual sensor data. Subsequently, we conducted a sensor-based performance analysis using all physiological data from each sensor. This involved concatenating each data type into a 1D sequence for sensor fusion, and training with our best model (Transformer). Model performance was evaluated using accuracy and weighted-average f1-score metrics (considering proportional contribution

of true instances of each class to the overall f1-score) on the test set through the Leave-One-Subject-Out (LOSO) approach to gain insights into the model’s learning behavior. Our Transformer model consistently outperformed LSTM and CNN-LSTM models, as shown in Table 3.

The ECG (Polar H10) data stood out as the most reliable indicator, achieving 82% accuracy and 80% f1-score in fear stimuli recognition in our study. The Transformer model’s self-attention mechanism demonstrated a remarkable ability to interpret heart rate variability (HRV), RR interval, and other parameters reflecting autonomic nervous system activity [38]. ACC (Empatica) emerged as the second-best indicator for fear recognition (0.77% acc, 0.76% f1), with bodily movements contributing to recognizing expressions of fear [23]. Player-specific movements, such as shooting and coping with enemies, exhibited distinct patterns enabling the models to identify fear.

Certain physiological data (TEMP, IBI, HR, EDA) exhibited lower performance compared to others. TEMP, in particular, showed the lowest success in detecting fear stimuli, aligning with results from Bruns et al. [7]. The low sampling rate of TEMP might make it challenging to capture temporal patterns of fear stimuli, resulting in a loss of temporal information [4]. In general, a 20-second time window provided more accurate results. This observation may be attributed to the volatility of emotions, as longer segments might cover fluctuating emotional states, making characterization more challenging. Additionally, longer segments would require more complex models, which are harder to train with limited labeled data. Shorter segments are faster to process, allowing for more training epochs and smaller learning rates. Hence, 20-second segments will be used in subsequent experiments.

Combining different data features for the best model (Transformer) with a 20-second time window significantly increased model performance to 84.4% f1-score (86.2% accuracy) for Polar H10 and 82.1% f1-score (82.9% accuracy) for Empatica. Confusion matrices for all three devices can be seen in Figure 3, revealing that the feed at medium fear levels was frequently confused with other classes, especially Eye-Tracking data confused with high fear levels.

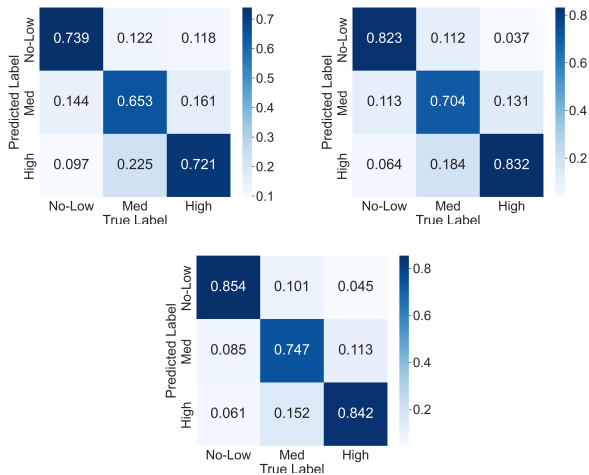
While numerous studies have delved into emotion recognition, such as those involving the DEAP [20] dataset and valence-arousal classification, there is a limited focus on fear classification. Table 4 presents some relevant studies on the topic. Our study, notably transformer model, using a Polar H10 sensory data fusion approach, outperforms similar ones using commercial games, focusing on two

Table 3: Accuracy and weighted-average f1-score results on the test set with LOSO for 10, 20 and 40-second time windows.

Physio. Data	10s				20s				40s			
	CNN-LSTM acc f1	LSTM acc f1	Bid-LSTM acc f1	Transf. acc f1	CNN-LSTM acc f1	LSTM acc f1	Bid-LSTM acc f1	Transf. acc f1	CNN-LSTM acc f1	LSTM acc f1	Bid-LSTM acc f1	Transf. acc f1
TEMP (E)	0.53 0.51	0.56 0.54	0.57 0.54	0.58 0.55	0.56 0.57	0.59 0.58	0.59 0.61	0.60 0.62	0.54 0.59	0.55 0.56	0.56 0.58	0.58 0.60
IBI (E)	0.57 0.56	0.57 0.58	0.60 0.58	0.63 0.62	0.60 0.59	0.62 0.60	0.64 0.66	0.67 0.66	0.59 0.58	0.62 0.60	0.63 0.62	0.65 0.60
HR (P)	0.59 0.56	0.62 0.59	0.64 0.63	0.67 0.62	0.64 0.64	0.68 0.61	0.68 0.65	0.70 0.67	0.62 0.65	0.64 0.65	0.68 0.63	0.68 0.62
EDA (E)	0.60 0.59	0.63 0.61	0.65 0.66	0.69 0.66	0.64 0.66	0.67 0.66	0.65 0.68	0.70 0.68	0.63 0.66	0.65 0.66	0.68 0.66	0.68 0.66
ACC (P)	0.71 0.66	0.71 0.64	0.73 0.66	0.72 0.66	0.72 0.66	0.71 0.66	0.75 0.66	0.76 0.71	0.72 0.72	0.75 0.73	0.75 0.70	0.72 0.68
Eye-Track. (H)	0.72 0.66	0.73 0.67	0.75 0.74	0.72 0.74	0.73 0.70	0.74 0.74	0.75 0.72	0.76 0.70	0.70 0.69	0.75 0.72	0.76 0.73	0.73 0.71
BVP (E)	0.72 0.70	0.73 0.71	0.74 0.69	0.73 0.72	0.73 0.66	0.74 0.68	0.75 0.71	0.76 0.75	0.72 0.69	0.74 0.68	0.73 0.74	0.74 0.72
ACC (E)	0.73 0.71	0.74 0.69	0.75 0.71	0.74 0.72	0.74 0.72	0.73 0.73	0.74 0.71	0.77 0.76	0.74 0.71	0.75 0.74	0.71 0.69	0.70 0.71
ECG (P)	0.76 0.73	0.77 0.75	0.78 0.77	0.79 0.79	0.76 0.75	0.77 0.73	0.80 0.78	0.82 0.80	0.74 0.73	0.74 0.75	0.76 0.73	0.79 0.76

Table 4: Fear classification accuracy results from the literature and comparison with our result

Work	Physio. Data	Methods	Best Acc.	Objective	VR Env.	Class	Subjects
Balan et al. [6]	EEG, EDA, HR	kNN,SVM,LDA	89.5/42.5%	Fear of Height	Self-Developed	2/4	8
Hu et al. [16]	EEG	CNN (VGG-16)	88.7%	Fear of Height	Richie's Plank	4	60
Luong et al.[22]	PPG, EDA, Respiration	LR,SVM, RF	71%	Fear	Commercial Game	2	24
Fu et al. [13]	Skeletal Points (Kinect)	Bid-LSTM	71.5%	Fear	Commercial Game	2	25
Our work	ECG, ACC (Polar H10)	Transformer	86.2%	Fear	Commercial Game	3	34

**Figure 3: Confusion matrices for for HMD, Empatica (top-left, top-right) and Polar H10 (bottom) using Transformer with 20 seconds time window, respectively.**

classes [13, 22]. Though our results slightly lag behind EEG studies, it's worth noting that EEG setups limit participants' movements more than the unobtrusive wearables we used and Polar H10 result shows that the sensory device is feasible for considerably robust for fear classification tasks and preferable instead of complex electrode-based devices like EEG. Additionally, solely studying the fear of heights [6, 16] makes the research less validated compared to studies exploring interactive environments like survival/horror games that include different dimensions of the fear.

Finally, creating a specific VR environment to induce fear may restrict participant movement and tasks. Commercial games, on the

other hand, provide participants with unstructured environments, allowing more freedom of movement and realistic actions. This characteristic makes commercial games valuable tools for validation experiments.

Limitations: Using a questionnaire for subjective feedback to obtain ground-truth data may not be the ideal method and comes with notable drawbacks. It requires active user feedback, potentially causing distraction and additional workload or breaking immersion and flow. Additionally, body motion data was not utilized in this study. Combining eye-tracking data with motion data from controllers and HMD might enhance classifier performance.

6 CONCLUSION

This study utilized a commercial survival-horror game for an exploratory analysis of fear responses in physiological data. We employed four from-scratch deep learning models for three levels of fear classification. ECG data, specifically from Polar H10, exhibited the most significant response with 82% accuracy and 80% f1-score for fear stimuli recognition. Furthermore, the sensor fusion approach with the Polar H10 (84.4% f1-score) device outperformed HMD and Empatica, demonstrating strong evidence for the device's applicability in fear recognition tasks. Future work will explore sensor fusion approaches with different state-of-the-art models for classifying fear levels.

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